SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

Agile Sprint Planning & User Story Workshop

Smart Industrial Worker Training and Certification Management System

CSA10 / CSA1005 – Software Engineering | CO1 Assessment Tool 2

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1. Project Vision and Stakeholders

1.1 Project Vision

For industrial organisations that must prove every worker holds valid safety, technical and compliance qualifications, the Smart Industrial Worker Training and Certification Management System is a centralised web application that maintains worker training profiles, schedules training, records attendance and assessments, tracks certificates, and warns managers before certifications expire. Unlike manual registers and spreadsheets, it prevents expired qualifications, exposes training gaps early and produces audit-ready compliance reports.

1.2 Objectives

- **Centralise:** Keep one accurate, searchable training profile for every employee.
- **Plan:** Manage courses, schedule sessions, enrol workers and track attendance.
- **Certify:** Record assessment results and certificates with issue and expiry dates.
- **Prevent:** Raise renewal alerts (for example 90, 60 and 30 days before expiry) and flag employees whose role needs a missing qualification.
- **Report:** Generate compliance reports for managers and auditors.

1.3 Stakeholders

Stakeholder	Role in the project	Key interest
Industrial workers	End users; attend training and hold certificates	View own records, upcoming sessions and renewal dates
Supervisors / department managers	Primary users; monitor their teams	Know who is non-compliant or about to expire; plan shift cover
Training coordinator / HR administrator	Product owner proxy; main operator	Create courses, schedule sessions, maintain employee records
Trainers / instructors	Deliver sessions and assessments	Mark attendance and enter assessment scores quickly
Safety and compliance officer	Business sponsor for compliance	Audit-ready reports; zero expired mandatory certificates
Plant management / senior leadership	Funds and approves the system	Reduced risk, fewer penalties, better training planning

Stakeholder	Role in the project	Key interest
External auditors / regulators	Consume reports	Reliable evidence of compliance
Development team, Scrum Master, IT admin	Build, run and support the system	Clear requirements, secure and maintainable system

2. Epics and User Stories

The requirements are divided into eight epics, which match the epics created in Jira (SIWTC-1 to SIWTC-8). Stories follow the format: As a <user>, I want <feature>, so that <benefit>.

Epic	Epic name	Summary
E1	Employee Profile Management	Register employees and maintain departments, roles and training history
E2	Training Program Management	Create courses, durations, validity periods and prerequisites
E3	Training Session Scheduling	Schedule sessions, assign trainers and enrol workers
E4	Attendance Management	Record attendance and course completion
E5	Assessment Management	Record scores, pass/fail results and reassessments
E6	Certification Management	Record certificates, issue and expiry dates, renewal status
E7	Expiry Alerts and Training Gap Analysis	Detect expiring or missing qualifications and notify managers
E8	Compliance Reports and Security	Role-based access and compliance reporting

ID	Epic	User story
US-01	E1	As an HR administrator, I want to register an employee with ID, department and job role, so that the organisation keeps accurate training records.
US-02	E8	As a user, I want to log in with a role-based account, so that I can only see and change the data my role allows.
US-03	E1	As a training coordinator, I want to view and update an employee's training history, so that I know which courses and certificates they hold.
US-04	E2	As a training coordinator, I want to create a training course with duration, validity period and mandatory flag, so that required training is standardised.
US-05	E1	As a supervisor, I want to search and filter employees by department and role, so that I can quickly review my team.
US-06	E3	As a training coordinator, I want to schedule a training session with a date, venue and trainer, so that workers can be trained on time.
US-07	E3	As a supervisor, I want to enrol workers in a session, so that every required worker is booked.

ID	Epic	User story
US-08	E4	As a trainer, I want to record who attended a session, so that completion can be verified.
US-09	E5	As a trainer, I want to record assessment scores with pass/fail status, so that certification decisions are based on results.
US-10	E6	As a training coordinator, I want to register a certificate with issue and expiry dates, so that its validity is tracked.
US-11	E7	As a manager, I want automatic alerts before a certificate expires, so that renewals are done in time.
US-12	E7	As a manager, I want to see employees missing a qualification required for their role, so that I can close training gaps.
US-13	E8	As a compliance officer, I want to generate a compliance report by department, so that I can show auditors our certification status.
US-14	E8	As a compliance officer, I want to export reports to PDF or Excel, so that I can share them outside the system.

3. Acceptance Criteria (Given–When–Then)

Criteria are written for the five stories selected for Sprint 1.

Story	Acceptance criteria
US-01 Register employee	1. Given the HR administrator is on the registration form, when they enter a unique employee ID, name, department and job role and click Save, then the record is stored and shown in the employee list within 2 seconds. 2. Given an employee ID already exists, when the administrator saves a new record with the same ID, then the system rejects it and shows the message "Employee ID already exists". 3. Given a required field is empty, when the administrator clicks Save, then the record is not saved and the empty field is highlighted.
US-02 Role-based login	1. Given a registered user, when they enter a correct username and password, then they reach the dashboard for their role (Admin, Supervisor, Trainer, Worker). 2. Given an incorrect password, when the user attempts login, then access is denied and after 5 failed attempts the account is locked. 3. Given a Worker is logged in, when they try to open the employee registration page, then access is denied.
US-03 Training history	1. Given an employee record exists, when the coordinator opens it, then the profile lists all courses, results and certificates with dates. 2. Given the coordinator adds a completed course, when they click Save, then the entry appears in the history immediately and the change is logged with the user name and time.

Story	Acceptance criteria
US-04 Create course	1. Given the coordinator enters a course name, duration (hours), validity (months) and mandatory flag, when they save, then the course appears in the course list. 2. Given a course name already exists, when the coordinator saves a duplicate, then the system rejects it. 3. Given a validity period of 0 or less, when the coordinator saves, then the system shows a validation error.
US-05 Search employees	1. Given 50 or more employee records, when the supervisor filters by department, then only employees of that department are listed. 2. Given a name or ID is typed in the search box, when the supervisor presses Search, then matching employees are returned within 3 seconds. 3. Given no record matches, when the search runs, then the message "No employees found" is displayed.

4. Product Backlog Prioritisation

Stories are ordered by business value (safety and compliance impact), customer need and dependencies: a story cannot be built until the data it relies on exists. Priority uses the MoSCoW scale.

Rank	Story	Priority	Depends on	Reason	Sprint
1	US-02 Role-based login	Must	–	Security foundation for every module	1
2	US-01 Register employee	Must	US-02	All records belong to an employee	1
3	US-04 Create course	Must	US-02	Needed before sessions and certificates	1
4	US-03 Training history	Must	US-01	Core training profile	1
5	US-05 Search employees	Should	US-01	Usability; supports managers	1
6	US-06 Schedule session	Must	US-04	Training planning	2
7	US-07 Enrol workers	Must	US-01, US-06	Links workers to sessions	2
8	US-08 Record attendance	Must	US-07	Proves completion	3
9	US-09 Record assessments	Must	US-08	Basis for certification	3
10	US-10 Register certificate	Must	US-09	Highest compliance value	4
11	US-11 Expiry alerts	Must	US-10	Directly prevents expired certificates	4
12	US-12 Training gaps	Should	US-10	Identifies missing qualifications	5

Rank	Story	Priority	Depends on	Reason	Sprint
13	US-13 Compliance report	Must	US-10	Audit evidence	5
14	US-14 Export reports	Could	US-13	Convenience	5

5. Sprint Backlog (Sprint 1)

Sprint 1 lasts two weeks. Stories US-02, US-01, US-04, US-03 and US-05 are selected because they are the highest priority and have no unfinished dependencies. Task hours are estimates.

Story	Development tasks	Hours
US-02 Login	• Design users and roles tables • Build login page and authentication • Implement role-based access and lock after 5 failures • Test login and permissions	14
US-01 Register employee	• Design employee table • Build registration form with validation • Reject duplicate IDs • Test form and validation	9
US-04 Create course	• Design course table • Build course form and list page • Validate duplicates and validity period	8
US-03 Training history	• Design training history table linked to employee • Build profile page showing history • Add audit log for changes	12
US-05 Search employees	• Build search and department filter • Add "No employees found" message • Test with sample data	5
Sprint tasks	• Set up repository and database • Prepare sample data • Sprint review demo and documentation	8
Total		56

6. Story Point Estimation

Technique: Planning Poker using the Fibonacci sequence (1, 2, 3, 5, 8, 13). US-01 (Register employee) is the reference story, a simple form with validation, and is set to 3 points. Other stories are sized relative to it.

Story	Points	Justification
US-01 Register employee	3	Reference story: one form, one table, simple validation.
US-02 Role-based login	5	Security-sensitive; roles, lockout and permissions need extra testing.

Story	Points	Justification
US-03 Training history	5	Links employees to courses and results; needs an audit log.
US-04 Create course	3	Similar in size to the reference story.
US-05 Search employees	2	Small; filtering on existing data, low risk.
Total	18	Within the assumed team velocity of 18–20 points per two-week sprint.

7. Sprint Goal and Expected Deliverables

Sprint 1 goal: Deliver a secure foundation of the system in which authorised users can log in, register employees, maintain their training history, define training courses and search for employees.

Deliverables at the end of Sprint 1

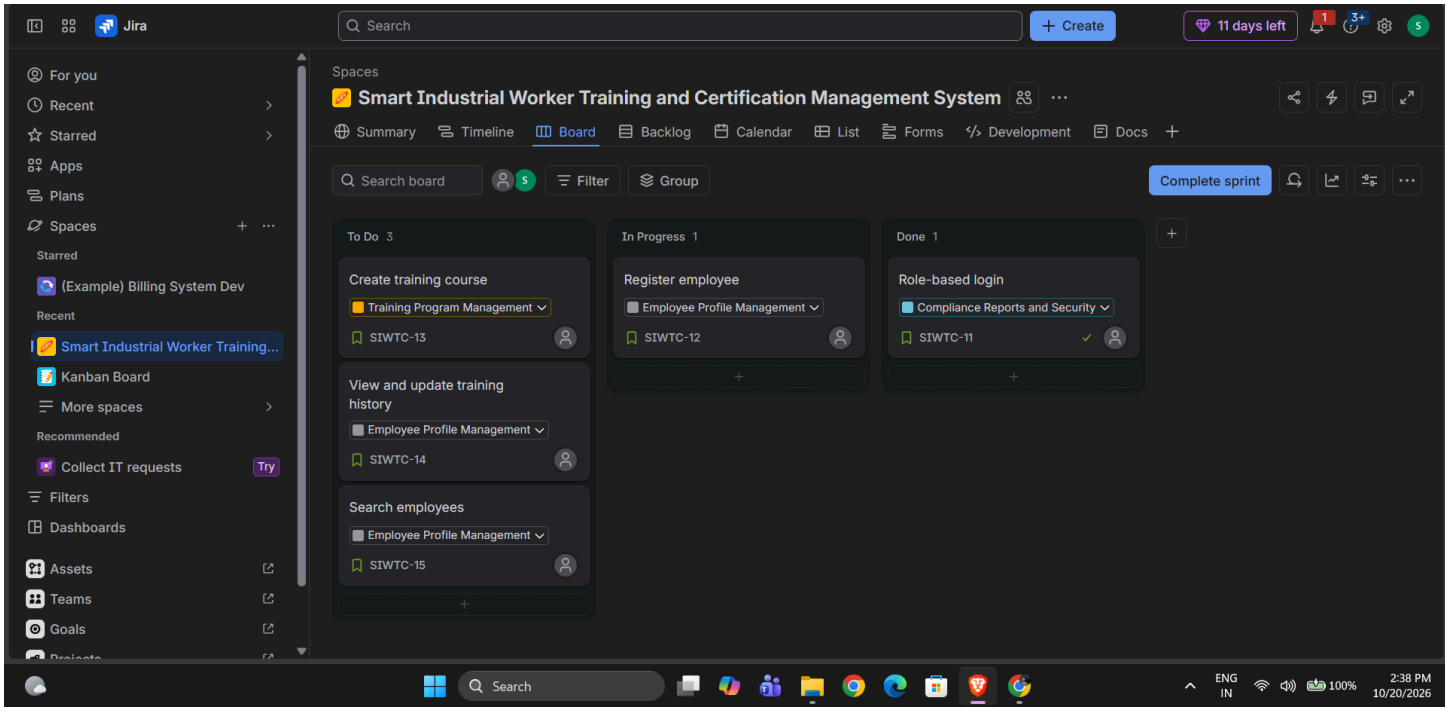
- Role-based login for Admin, Supervisor, Trainer and Worker.
- Employee registration with validation and duplicate-ID protection.
- Employee profile showing training history.
- Course creation and course list.
- Employee search and department filter.
- Tested increment, sample data, and updated Jira board and sprint report.

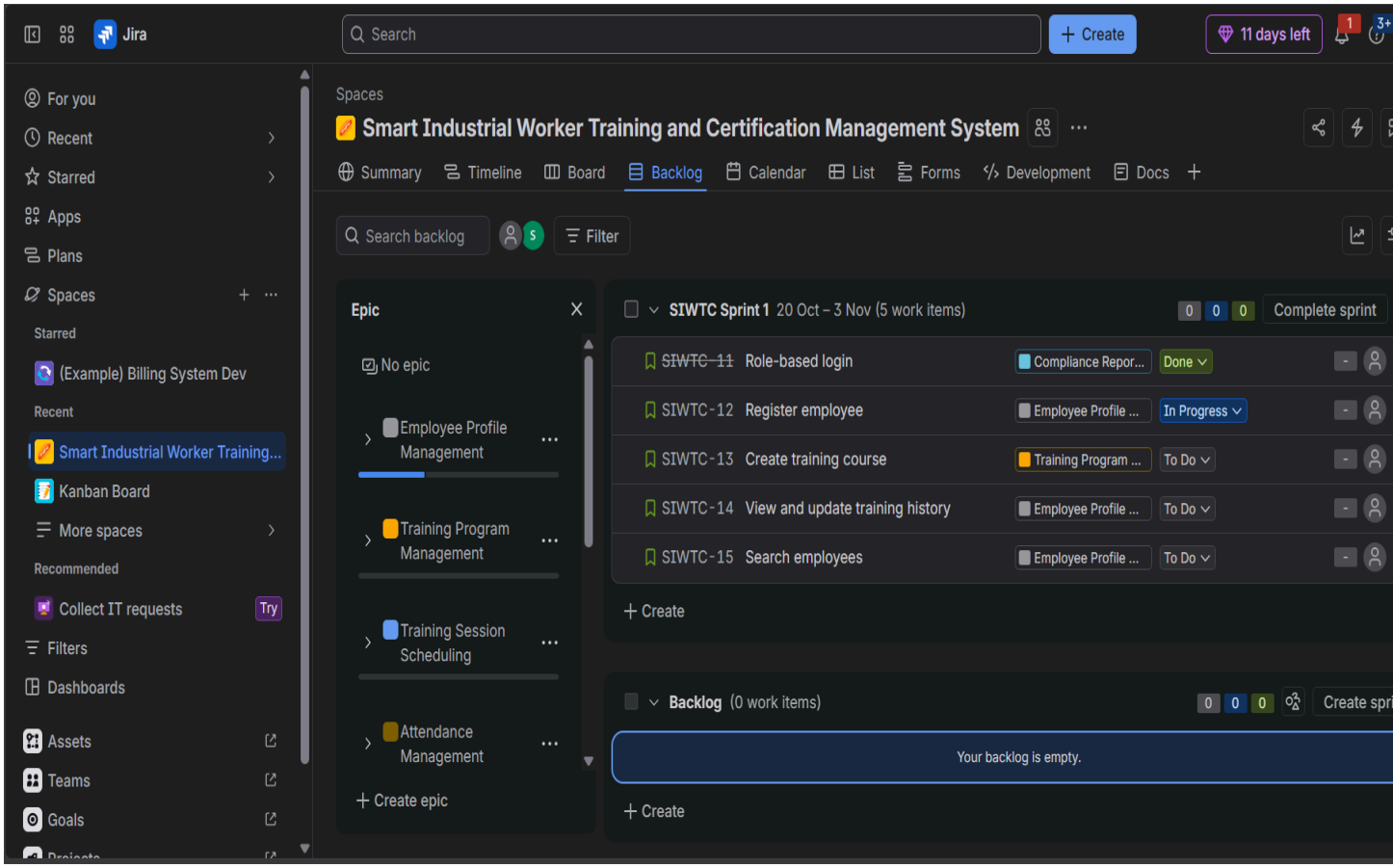
Definition of Done

A story is done when its acceptance criteria pass, the code is reviewed and tested, no critical defects remain, and the item is moved to Done on the Jira board.

Jira evidence to attach

Screenshots from the SIWTC Jira space: (1) epic list showing the eight epics, (2) backlog with stories assigned to Sprint 1, (3) one story with its acceptance criteria, and (4) the Sprint 1 board.




Jira
Search
Create
11 days left
For you
Recent
Starred
Apps
Plans
Spaces
Starred
(Example) Billing System Dev
Recent
Smart Industrial Worker Training...
Kanban Board
More spaces
Recommended
Collect IT requests
Try
Filters
Dashboards
Assets
Teams
Goals
Spaces
Smart Industrial Worker Training and Certification Management System
Summary
Timeline
Board
Backlog
Calendar
List
Forms
Development
Docs
Search backlog
Filter
Epic
No epic
Employee Profile Management
Training Program Management
Training Session Scheduling
Attendance Management
Create epic
SIWTC Sprint 1 20 Oct – 3 Nov (5 work items)
Complete sprint
SIWTC-11 Role-based login
Compliance Repor...
Done
SIWTC-12 Register employee
Employee Profile ...
In Progress
SIWTC-13 Create training course
Training Program ...
To Do
SIWTC-14 View and update training history
Employee Profile ...
To Do
SIWTC-15 Search employees
Employee Profile ...
To Do
Create
Backlog (0 work items)
Your backlog is empty.
Create